

# Analysis and control of the thermal runaway of ceramic slab under microwave heating

LIU ChangJun<sup>1†</sup> & Dongwoo SHEEN<sup>2</sup>

<sup>1</sup> School of Electronics & Information Engineering, Sichuan University, Chengdu 610064, China;

<sup>2</sup> Department of Mathematics, Seoul National University, Seoul 151-747, Korea

Thermal runaway is a special macroscopic phenomenon of the dielectrics during microwave heating, in which there is a big jump of the steady state temperature while the applied microwave power varies slightly. It hinders the applications of microwave heating technique in industry. A simulation based on the finite difference time domain (FDTD) method to solve Maxwell's equations coupled with the finite difference (FD) method to solve a heat transfer equation (HTE) is presented, and the temperature variation in a ceramic slab during microwave heating is obtained. The temperature variation in the ceramic slab during microwave heating is simulated with various ceramic parameters and applied microwave powers so as to analyze the condition under which thermal runaway is introduced. Moreover, a microwave power control method, based on a single temperature threshold and dual applied microwave powers, is presented, which improves microwave heating efficiency and controls thermal runaway. The relation between the final applied microwave power and the monitored temperature threshold is presented as well. This method can be applied in many fields related with microwave heating techniques.

finite difference time domain method (FDTD), heat transfer, temperature, power threshold

## 1 Introduction

Microwave heating can speed up material processing of many applications in various fields, such as chemical industry, pharmaceuticals industry, food industry, mining industry and so on<sup>[1–6]</sup>. During microwave heating, the material characteristics such as the conductivity, permittivity, and heat transfer constant vary with the temperature rise<sup>[7–10]</sup>. It may lead to some special macroscopic thermal phenomenon. When microwave heating is applied to ceramic sintering<sup>[11]</sup>, mineral grinding<sup>[6]</sup>, chemistry and pharmaceuticals<sup>[1,2,12]</sup>, rubber processing, and so on, the local “hot spot” might appear<sup>[13]</sup>, there may occur a very high temperature rise, which usually leads to the phenomenon of thermal runaway.

Received March 22, 2007; accepted December 3, 2007

doi: 10.1007/s11431-008-0221-7

<sup>†</sup>Corresponding author (email: [cjliu@ieee.org](mailto:cjliu@ieee.org))

Supported by the National Natural Science Foundation of China (Grant Nos. 60301004 and 60411140521), Korea Science and Engineering Foundation (Grant Nos. F01-2004-000-10263-0, R14-2003-019-01000-0 and KRF-2006-070-C00014)

Thermal runaway is an unstable macroscopic phenomenon during microwave heating, which can be basically described in terms of: a) the steady state temperature, b) the temperature rise rate<sup>[14,15]</sup>. If there is a slight variation of either the applied microwave power or the parameters of the heated dielectrics, there is an enormous steady state temperature jump, which describes the steady state temperature thermal runaway. On the other hand, there occurs a very high temperature rise rate for a very short period of microwave heating. This describes the thermal runaway in terms of the temperature rise rate. In the process of microwave heating on ceramics, the dependency of either electric conductivity or thermal conductivity on temperature may lead to thermal runaway<sup>[9,14]</sup>.

In 1991 Kenker et al.<sup>[16]</sup> presented empirical formula on thermal runaway to study microwave heating on ceramics. Since Kriegsmann<sup>[14]</sup> analyzed a model of thermal runaway using an asymptotic method in 1992, many researchers have been attracted to discuss mathematical analysis of thermal runaway aspects<sup>[15,17,18]</sup>. In the numerical simulation of thermal runaway, a coupled system of Maxwell's equations and the heat transfer equation (HTE) has to be solved. In 1999, Alpert and Jerby<sup>[19]</sup> presented a coupling method with two time scales to solve it. Later, a finite element method with an adaptive time step scheme was developed<sup>[20]</sup>. Some applications of the FDTD method on microwave heating can be found in refs. <sup>[21–24]</sup>.

Concerning the control of thermal runaway, Kriegsmann<sup>[14]</sup> first presented an asymptotical analysis and the control on microwave power to avoid thermal runaway. James and John<sup>[25]</sup> presented a control method by the initial temperature and the time varying boundary condition, and performed mathematical analysis. It seems that there are many limitations to realize those methods in industry. A practical method is to control microwave power with a temperature feedback system<sup>[10,26]</sup>. However, the continuous microwave power adjustment and the detection of the temperature time gradient are complex, difficult and not suitable for industry application.

The aim of the present paper is twofold: we present a new numerical method based on the FDTD method to solve the coupled system of Maxwell's equations and the HTE so as to perform simulation of a ceramic slab under microwave heating. The steady state temperature distribution of the ceramic slab that is dependent on the applied microwave power is achieved with various ceramic slab parameters. Meanwhile, the temperature rise in the ceramic slab is obtained by the simulation method. Furthermore, a power controlling method to avoid thermal runaway during microwave is discussed. A relation between the final state microwave power and the monitored temperature threshold is shown. The simulation method presented here enables to simulate more complex microwave heating model and to promote the development of microwave heating research and applications.

## 2 Microwave heating model on ceramics

The microwave heating on a ceramic slab in transverse electromagnetic (TEM) mode is modeled following refs. <sup>[14, 27, 18]</sup>, as shown in Figure 1. The ceramic slab with length  $l$  and relative dielectric constant  $\varepsilon_r$  is assumed to be exposed to the free space. The microwave with the  $x$ - and  $y$ -components of the electromagnetic fields, respectively, is assumed to propagate at frequency 2450 MHz along the  $z$ -direction. The electromagnetic fields then satisfy Maxwell's equations:

$$\begin{cases} \frac{\partial E_x}{\partial z} = \mu_r \mu_0 \frac{\partial H_y}{\partial t}, \\ \frac{\partial H_y}{\partial z} = \varepsilon_r \varepsilon_0 \frac{\partial E_x}{\partial t} + \sigma E_x. \end{cases} \quad (1)$$

Since the ceramic slab is a lossy material, it absorbs microwave power and leads to temperature rise. The temperature variation is governed by the HTE, i.e.,

$$\rho C_p \frac{\partial T(z,t)}{\partial t} = K \frac{\partial^2 T(z,t)}{\partial z^2} + P_d(z,t), \quad (2)$$

where  $T$ ,  $C_p$ ,  $K$  and  $P_d$  are the temperature, ceramic density, thermal capacitance, thermal conductivity, and microwave power dissipation, respectively. If the polarization loss is not considered, the microwave power dissipation is given by  $P_d(z,t) = \frac{1}{2} \sigma(z,t) |E_x(z,t)|^2$ . Assume that the ambient temperature is kept at  $T_0$ . The boundary condition of the HTE is given by

$$K \frac{\partial T(z,t)}{\partial z} = h[T(z,t) - T_0] + se[T^4(z,t) - T_0^4], \quad (3)$$

where  $h$  and  $e$  are the convective heat and radiation heat constants, respectively, and  $\sigma = 5.67 \times 10^{-8} \text{ kg/s}^3/\text{K}^4$  denotes the Stefan-Boltzmann constant. The conductivity increases with the rise of temperature, which can be described by an exponential or Arrhenius model<sup>[14,18]</sup>:

$$\sigma(T) = \sigma_0 \exp\left(\frac{T - T_0}{T}\right), \quad (4)$$

where  $\sigma_0 = 1.63 \times 10^{-3} \text{ S/m}$  is the ceramic conductivity at normal temperature  $T_0 = 300 \text{ K}$ .

### 3 Simulation method

Since the dielectric characteristics of the ceramics vary with temperature rise under microwave heating, the time varying medium has to be considered during electromagnetic simulation. A standard FDTD method is adopted to solve Maxwells equations so as to couple the HTE in dealing with time varying medium<sup>[21]</sup>. The FDTD simulation system is shown in Figure 1. Outside the ceramic slab the 1st order Mur absorbing boundary conditions (ABC) are imposed at  $z = -a$  and  $z = a$ . The connection boundary condition (CBC) is then located between the ceramic slab and the left ABC. The time step  $\Delta t$  and the space step  $\Delta z$  in the FDTD method are constrained by the CFL (Courant-Friedrichs-Lewy) condition. Here the time step is set as  $\Delta t = \frac{c}{2\Delta z}$ , with  $c$  being the speed of light in vacuum. The finite difference method (FDM) with forward Euler scheme is adopted to solve eq. (2) as follows:

$$T^{n+\frac{1}{2}}(k) = T^{n-\frac{1}{2}}(k) + \frac{\Delta t_{\text{HTE}}}{\rho C_p} \left[ K \frac{T^{n-\frac{1}{2}}(k+1) - 2T^{n-\frac{1}{2}}(k) + T^{n-\frac{1}{2}}(k-1)}{\Delta z^2} + \frac{1}{2} \sigma^{n-\frac{1}{2}}(k) |E^n(k)|^2 \right], \quad (5)$$

where  $\Delta t_{\text{HTE}}$  is the time step,  $k$  represents the spatial position  $z = k\Delta z$ ,  $\Delta z = l/N_k$ , and the superscript

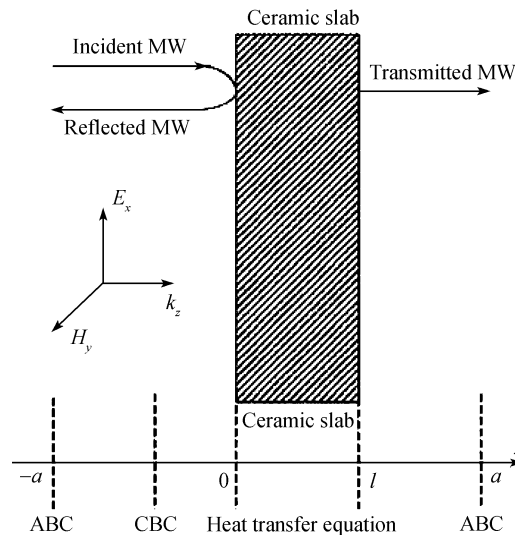
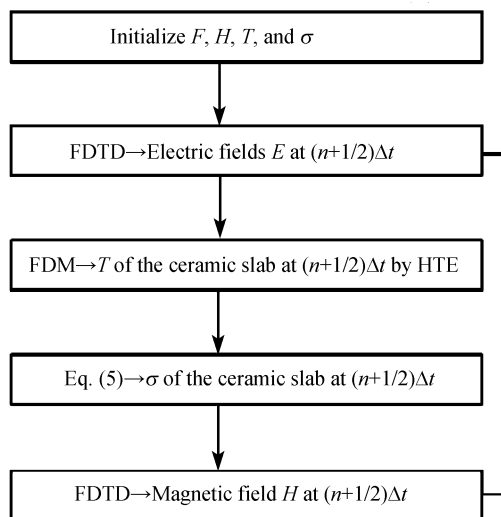


Figure 1 Model of microwave heating on a ceramic slab.

denotes  $t=nt_{\text{HTE}}$ . The stability condition for eq. (5) is given by  $\Delta t_{\text{HTE}} < \frac{\rho C_p}{2K} \Delta z^2$ . After updating the values of  $T^{n+1/2}(k)$  for  $k = 1 \dots N_{k-1}$  using eq. (5), the value of  $T^{n+1/2}(0)$  is updated by the central difference approximation of the spatial derivative of eq. (3) as follows:

$$T^{n+\frac{1}{2}}(1) = T^{n-\frac{1}{2}}(1) + \frac{\Delta t_{\text{HTE}}}{\rho C_p} \left[ K \frac{T^{n-\frac{1}{2}}(2) - T^{n-\frac{1}{2}}(1)}{\Delta z^2} - h \left( T^{n-\frac{1}{2}}(1) - T_0 \right) - se \left( T^{n-\frac{1}{2}}(1)^4 - T_0^4 \right) + \frac{1}{2} \sigma^{n-\frac{1}{2}}(1) |E^n(1)|^2 \right]. \quad (6)$$

With respect to microwave at 2450 MHz, the time step in the FDTD method is typically at the



**Figure 2** Flow chart of the coupling method.

order of  $10^{-11}$  s, while the microwave heating duration normally lasts at the order of  $10^2$  s. If the conventional FDTD method is directly applied, there are at least  $10^{13}$  steps to compute. To speed up the simulation, the time factor  $\alpha$ , which is  $10^{10}$  is applied to transforming the time step in the FDTD and HTE into the same order<sup>[21,23,24]</sup>. Figure 1 shows the method to couple Maxwell's equations and the HTE together, in which the electric fields, the magnetic fields, the temperature and the conductivity are updated in turn. The domain to solve the HTE is  $[0, l]$ , and the domain to apply the FDTD is  $[-a, a]$ . Based on the model and the coupling method shown in Figures 1 and 2, the simulation of microwave heating on the ceramic slab can be performed.

## 4 Simulation results and analysis

The electric field amplitude of the incident microwave is  $E_0$ . The dependence of the ceramic permittivity and thermal conductivity on temperature was neglected in the simulation. Other ceramic parameters related with the HTE were obtained from ref. [19]. Some main parameters in simulation are shown in Table 1.

**Table 1** Parameters in FDTD and HTE

| $t_{\text{FDTD}}$ (ps) | $Z$ (mm) | $t_{\text{HTE}}$ (ms) | $\varepsilon_r$ | $K$ (W/m/K) | $e$ | $H$ (W/K) |
|------------------------|----------|-----------------------|-----------------|-------------|-----|-----------|
| 0.504                  | 0.302    | 5.04                  | 10              | 10          | 0.7 | 170       |

### 4.1 Microwave power threshold of thermal runaway

The steady state temperature dependent on microwave power shows the system response of the long duration microwave heating. The thickness of the ceramic slab is 4 cm, and the relative temperature is defined as  $\theta = (T - T_0)/T_0$ . When the initial temperature of the ceramic slab is of

$\theta_0 = 0$  and  $\theta_0 = 7$ , respectively, the dependence of the steady state temperature on microwave power is simulated.  $\theta_0 = 7$  shows a cooling down procedure, when the absorbed microwave power is less than the heat loss rate. The central temperature  $\theta_C$  of the ceramic slab is shown in section 4.2, where the applied microwave power density is given by  $\frac{1}{2}\omega\epsilon_0\epsilon_r E_0^2$  (also called microwave power in the paper). The simulated results agree well with those in ref. [18].

If the initial temperature of the ceramic slab is low, the steady state temperature at the center jumps from the lower branch to the higher branch once the applied microwave power is higher than  $P_{TH}$ . In the opposite condition, the applied microwave power is judged by  $P_{TL}$ . The applied microwave powers  $P_{TL}$  and  $P_{TH}$  are defined as the microwave power thresholds of thermal runaway. At low initial temperature, the slight variation around  $P_{TH}$  leads to enormous steady state temperature change of the ceramic slab, which is the phenomenon of thermal runaway.

#### 4.2 Thermal runaway procedure in time domain

When the initial temperature of the ceramic slab is  $\theta_0 = 0$  two microwave powers are applied in order to simulate the temperature rise in the time domain. Figure 4(a) indicates the temperature rise while the applied microwave power is  $P_a = 0.15 \times 10^{10} \text{ W/m}^3$  ( $P_a \approx P_{TH}$ , but  $P_a$  is slightly less than  $P_{TH}$ ). The temperature of the ceramic slab rises evenly while the rate of temperature rise drops. It takes about 270 min to reach a steady state. The highest temperature is located at the center of the ceramic slab, and the highest relative temperature is  $\theta_{max} = 1.1$ . There is no thermal runaway phenomenon in this case.

Figure 4(b) shows the temperature rise with the applied microwave power  $P_b = 0.16 \times 10^{10} \text{ W/m}^3$  ( $P_b \approx P_{TH}$ ,  $P_b$  being slightly greater than  $P_{TH}$ ). In the beginning stage, the temperature rises smoothly and the highest temperature is located at the center of the ceramic slab. At about 140 min of microwave heating, the temperature rises abruptly, and the temperature rise rate goes as high as 326 K/min. The phenomenon of thermal runaway appears. At about 200 min, the temperature goes to a steady state, while the highest relative temperature is  $\theta_{max} = 7.0$  at the position near  $z = 0$ .

Around the microwave power threshold  $P_{TH}$ , a slight power disturbance will lead to a big steady state temperature jump. Comparing Figures 4(a) and 4(b), the applied microwave power increase only by 6% entails into the highest steady state relative temperature increase by about 7 times, due to a strong nonlinearity. Furthermore, the closer the applied power is to the power threshold  $P_{TH}$ , the longer time is required to reach the steady state. Once the applied microwave power is higher than the power threshold  $P_{TH}$ , there will be a steep temperature rise.

#### 4.3 Effects on thermal runaway from ceramic slab parameters

The material parameters, such as dielectric constant, heat conductivity, and conductivity dependent

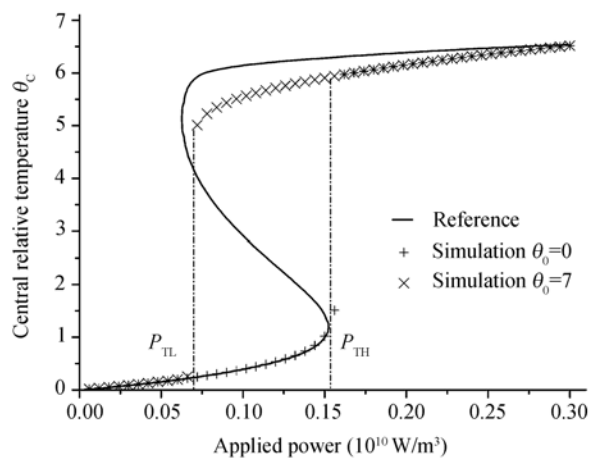


Figure 3 S-curve: Steady state central temperature.

on temperature, may change the microwave power threshold of thermal runaway. The geometric parameter (the thickness of the slab) plays a dominant role in the microwave power threshold. When the thickness of the ceramic slab varies, the central temperature of the steady state relative temperature dependent on the applied microwave power is shown in Figure 5. If the thickness of the slab is about 4 or 6 cm, a lower microwave power will lead to the phenomenon of thermal runaway. If the thickness is about 2.7 and 4.7 cm, much higher microwave power is required to lead to thermal runaway.

The steady state temperature is sensitive to the thickness of the ceramic slab at certain condition. For example, with the same applied microwave power, i.e.,  $0.2 \times 10^{10} \text{ W/m}^3$ , a slight thickness variation of 1 mm may lead to a huge steady state temperature jump. In the industrial application of microwave heating technique, if there is a slight difference in material under microwave heating, it may result in huge difference in microwave heating effects. If the parameters, such as dielectric constant and the heat conductivity, are changed, similar results may be obtained; a slight variation of one parameter may change the microwave threshold of thermal runaway significantly. Thus, in order to keep the microwave heating system working smoothly, the system should be designed to work in a stable region, in which the temperature is not too sensitive to ceramic parameters.

#### 4.4 The control of thermal runaway

The microwave heating efficiency is very low, especially when the applied microwave power is near  $P_{TH}$ . A microwave power control method is discussed based on a single temperature judgment and dual microwave power, which can improve microwave-heating efficiency and avoid thermal runaway. Firstly, a high microwave power  $P_H$  ( $P_H > P_{TH}$ ) is applied, recalling that  $P_{TH}$  and  $P_{TL}$  are the lower and higher power thresholds of thermal runaway. When the monitored relative temperature on the surface of the ceramic slab is higher than the preset temperature threshold  $\theta_T$ , the applied microwave power will be decreased to the final microwave power  $P_F$  ( $P_{TL} < P_F < P_{TH}$ ), and then it will be maintained to heat the ceramic slab at the desired stable state temperature.

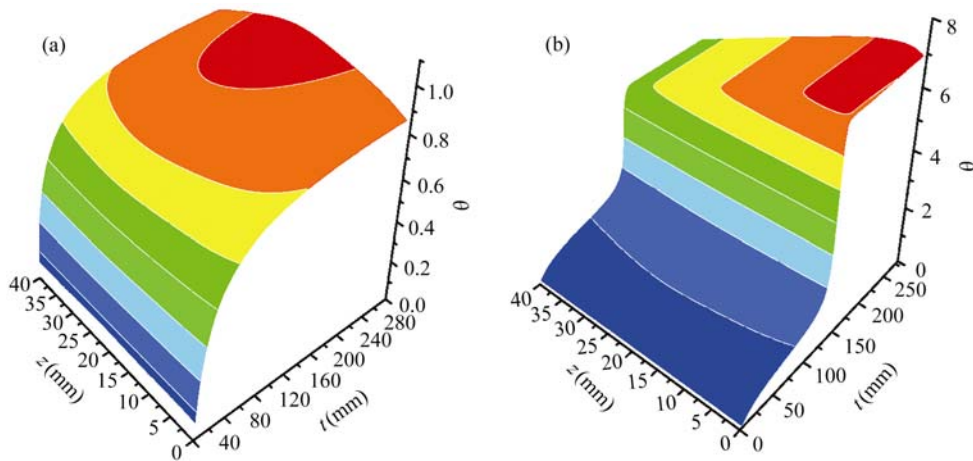
For example, in order to heat a ceramic slab of 4 cm thickness, initially the microwave power  $P_H = 0.5 \times 10^{10} \text{ W/m}^3$  is applied. When the monitored temperature arrives at  $\theta_T = 2.0$ , the applied microwave power is switched to  $P_F = 0.15 \times 10^{10} \text{ W/m}^3$ . During the 25 min of microwave heating, the temperature rise is shown in Figure 6(a). Notice that even with the final applied power  $P_F = 0.15 \times 10^{10} \text{ W/m}^3$  that is still lower than the microwave threshold  $P_{TH}$ , thermal runaway occurs.

In a further numerical simulation with the preset temperature threshold  $\theta_T = 1.0$ , the temperature rise is shown in Figure 6(b), in which there is no thermal runaway phenomenon. It takes much shorter time to reach the heat equilibrium state. A reasonable value of the preset temperature threshold corresponding to the final applied microwave power will shorten the microwave heating time significantly.

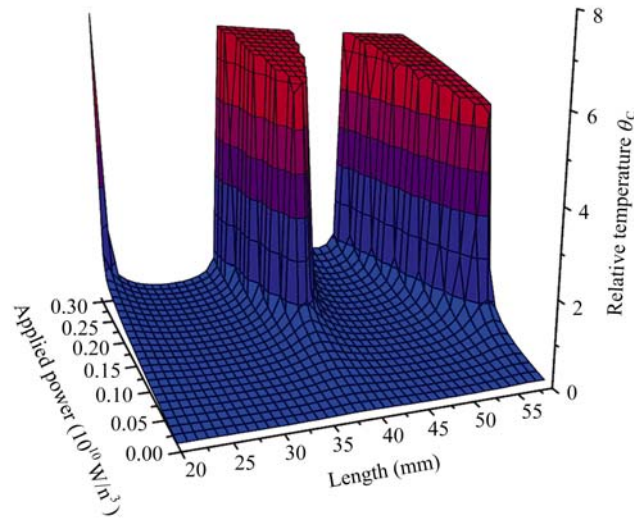
To avoid the phenomenon of thermal runaway, the preset temperature threshold  $\theta_T$  and the final applied microwave power  $P_F$  are given by

$$\theta_T \leq \theta(P_{TH}) + \frac{\theta(P_{TH}) - \theta(P_{TL})}{P_{TH} - P_{TL}}(P_F - P_{TH}), \quad (7)$$

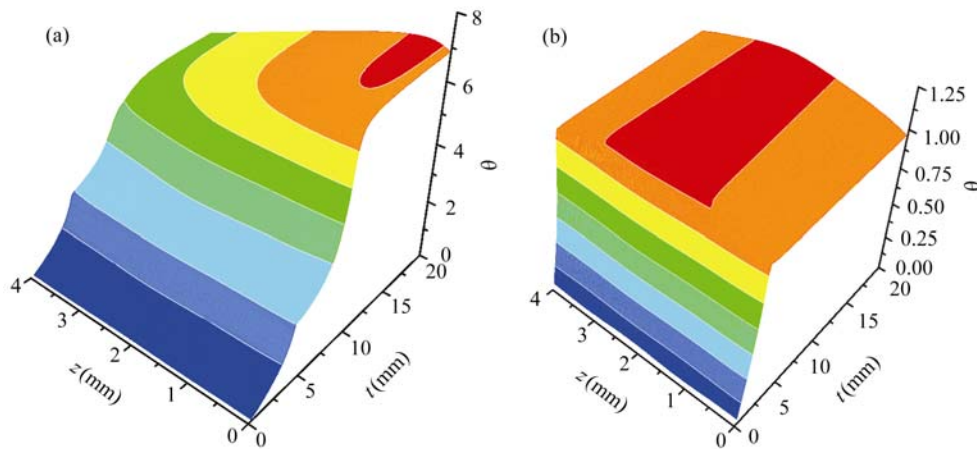
where  $\theta(P_{TH})$  and  $\theta(P_{TL})$  are the steady state relative temperature corresponding to microwave thresholds  $P_{TL}$  and  $P_{TH}$  in the lower and upper branches, respectively (see Figure 3). In order to improve the system safety, a reasonable lower temperature threshold  $\theta_T$  may be adopted. Since



**Figure 4** Time domain temperature rise in the ceramic slab. (a)  $P_a = 0.15 \times 10^{10} \text{ W/m}^3$ ;  $P_a \ll P_{TH}$ ; (b)  $P_b = 0.16 \times 10^{10} \text{ W/m}^3$ ;  $P_b \gg P_{TH}$ .



**Figure 5** Steady state temperature with the ceramic slab thickness.



**Figure 6** The control on thermal runaway. (a)  $P_r = 0.15 \times 10^{10} \text{ W/m}^3$ ,  $\theta_i = 2.0$ ; (b)  $P_r = 0.15 \times 10^{10} \text{ W/m}^3$ ,  $\theta_i = 1.0$ .

there are only two microwave power levels as well as one temperature threshold monitor in the system, the method of thermal runaway control is convenient, stable and easy to build.

## 5 Conclusion and discussion

The coupling method based on the FDTD method to solve Maxwell's equations and the FD method to solve HTE is applied to simulate thermal runaway during microwave heating on a ceramic slab. The strong nonlinearity and complexity of the microwave heating process are achieved. Based on the simulation and analysis, the conditions of thermal runaway are: 1) there is a dependence of material characteristics on temperature; 2) the applied microwave power is greater than a threshold. The microwave power threshold of thermal runaway is not only dependent on material characteristic, but also sensitive to the geometric parameters under certain conditions. If the applied microwave power is higher than the microwave power threshold, there is a steep temperature rise during microwave heating; otherwise, the temperature rises smoothly.

Under certain conditions, the microwave power of thermal runaway is sensitive to some parameters of the material under heating. For example, in the microwave heating on foods, it was found that the temperature rise and distribution were sensitive to the size and shape of the food<sup>[5]</sup>. A slight variation of the ceramic slab thickness may lead to a significant temperature jump of the steady state temperature from the simulation results. The sensitivity cannot be explained by the resonance between the applied microwave and the dielectrics under heating, but by the macroscopic behavior from the bifurcation of the nonlinear model.

A method of microwave power control based on single temperature threshold and dual temperature levels is presented in this paper, which can avoid the phenomenon of thermal runaway. The high microwave power is applied at the initial stage to speed up the heating process, while the final microwave power is applied to keeping the material in heat equilibrium. Thermal runaway is related not only to the applied microwave power but also to the current material temperature. A formula to estimate the reasonable final microwave power and the temperature threshold is given.

With the rapid development of microwave techniques, there are many new fields for microwave applications, such as microwave soldering<sup>[28]</sup>, microwave hybrid heating<sup>[29]</sup>, microwave thawing<sup>[30]</sup>, microwave drilling<sup>[31]</sup>, and so on. The Maxwell's equations coupled with the HTE are required to solve to analyze the heating process. The method presented in this paper is applicable in both thermal runaway control and microwave heating applications.

- 1 Oliver C K, Doris D. The impact of microwave synthesis on drug discovery. *Nature Revs Drug Disc*, 2006, 5(1): 51–63 [\[DOI\]](#)
- 2 Adam D. Microwave chemistry: Out of the kitchen. *Nature*, 2003, 421(6923): 571–572 [\[DOI\]](#)
- 3 Jerby E, Dikhtyar V, Aktushev O, et al. The microwave drill. *Science*, 2002, 298(5593): 587–589 [\[DOI\]](#)
- 4 Osepchuk J M. Microwave power applications. *IEEE Trans MTT*, 2002, 59(3): 975–985
- 5 Oliverira M E, Franca A S. Microwave heating of foodstuffs. *J Food Eng*, 2002, 53(4): 347–359 [\[DOI\]](#)
- 6 Walkiewicz J W, Clark A E, McGill S L. Microwave assisted grinding. *IEEE Trans Indust Appl*, 1991, 27(2): 239–242 [\[DOI\]](#)
- 7 Wu X, Thomas J R Jr. Control of thermal runaway in microwave resonant cavities. *J Appl Phys*, 2002, 92(6): 3374–3380 [\[DOI\]](#)
- 8 Nelson S O, Bartley P G Jr. Measuring frequency and temperature-dependent permittivities of food materials. *IEEE Trans Instrum Measur*, 2002, 51(4): 589–692 [\[DOI\]](#)
- 9 Parris P E, Kenker V M. Thermal runaway in ceramics arising from the temperature dependent of the thermal conductivity.

- Phys Stat, Sol (b), 1997, 200(1): 39–47 [\[DOI\]](#)
- 10 Bale G O, Li M. Robust temperature control for microwave heating of ceramics. IEEE Trans Indust Electr, 1997, 44(1): 124–131 [\[DOI\]](#)
  - 11 Chatterjee A, Basak T, Ayappa KG. Analysis of microwave sintering of ceramics. AIChE J, 1998, 44(10): 2302–2311 [\[DOI\]](#)
  - 12 McIntosh RD, Nolan P. Review of the selection and design of mitigation systems for runaway chemical reactions. J Loss Prev Proc Indust, 2001, 14(1): 7–42
  - 13 Zhang Y, Zhu S, Marchant T R. A DRBEM model for microwave heating problems. Appl Math Model, 1995, 19(5): 287–297 [\[DOI\]](#)
  - 14 Kriegsmann G A. Thermal runaway in microwave heated ceramics: a one-dimensional model. J Appl Phys, 1992, 71(4): 1960–1966 [\[DOI\]](#)
  - 15 Hill J M, Marchant T R. Modeling microwave heating. Appl Math Model, 1996, 20(1): 3–15 [\[DOI\]](#)
  - 16 Kenker V M, Skala L, Weiser M W, et al. Theory of microwave interactions in ceramic materials: the phenomenon of thermal runaway. J Mater Sci, 1991, 26(9): 2483–2489 [\[DOI\]](#)
  - 17 Vriezinger C A, Sanchez-Pedreno S, Grasman J. Thermal runaway in microwave heating—a mathematical analysis. Appl Math Model, 2002, 26(9): 1029–1038 [\[DOI\]](#)
  - 18 Gupta N, Midha V, Balakotaiah V, et al. Bifurcation analysis of thermal runaway in microwave heating of ceramics. J Electr Soc, 1999, 146(12): 4659–4665 [\[DOI\]](#)
  - 19 Alpert Y, Jerby E. Coupled thermal electromagnetic model for microwave heating of temperature dependent dielectric media. IEEE Trans PS, 1999, 27(2): 555–562
  - 20 Rabello A A, Silva E J, Saldanha R R, et al. Adaptive time-stepping analysis of nonlinear microwave heating problems. IEEE Trans Magnet, 2005, 41(5): 1584–1587 [\[DOI\]](#)
  - 21 Torres F C, Jecko B. Complete FDTD analysis of microwave heating processes in frequency-dependent and temperature-dependent media. IEEE Trans MTT, 1997, 45(1): 108–117 [\[DOI\]](#)
  - 22 Iskander M F. FDTD simulation of microwave sintering of ceramics in multimode cavities. IEEE Trans MTT, 1994, 42(5): 793–800 [\[DOI\]](#)
  - 23 Yan L, Huang K, Liu C, et al. Study on time scaling factor in the FDTD simulation of microwave heating. J Sichuan Univ (Nat Sci Ed), 2003, 40(2): 395–398
  - 24 Liu C, Huang K, Hu Z, et al. Experimental study and numerical simulation on the microwave nonlinear response due to chemical reaction. J Microwaves, 2000, 16(2): 165–171
  - 25 James G, John F. Control of thermal runaway—some mathematical insights. Int J Heat Mass Transfer, 1998, 41(19): 2979–2990 [\[DOI\]](#)
  - 26 Groombridge P, Oloyed A, Siores E. A control system for microwave processing of materials. J Manufact Sci Eng, 2000, 122(1): 253–261 [\[DOI\]](#)
  - 27 Vriezinger C A. Thermal profiles and thermal runaway in microwave heated slabs. J Appl Phys, 1999, 85(7): 3774–3779 [\[DOI\]](#)
  - 28 Kriegsmann G A, Luke J H C. Microwave joining of two long hollow tubes—an asymptotic theory and numerical simulations. J Eng Math, 2001, 39(1): 63–78 [\[DOI\]](#)
  - 29 Haala J, Wiesbeck W. Modeling microwave and hybrid heating processes including heat radiation effects. IEEE Trans MTT, 2002, 50(5): 1346–1354 [\[DOI\]](#)
  - 30 Lee M Z C, Marchant T R. Microwave thawing of cylinders. Appl Math Model, 2004, 28(8): 711–733 [\[DOI\]](#)
  - 31 Jerby E, Aktushev O, Dikhtyar V. Theoretical analysis of the microwave-drill near-field localized heating effect. J Appl Phys, 2005, 97(034909): 1–7 [\[DOI\]](#)